EE6347 - Devices for AI and Neuromorphic Computing

Assignment 4

Task 1: Passive RRAM Array

a. Create 3 x 3 RRAM Crossbar using the same model used in the Mid-Sem exam. (1 Mark)

b. Follow the steps we followed in Tutorial 5. Use a V/3 scheme. Write the (0,0) cell and read currents through (1,0) and (2,2) (2 Marks = 1+1)

c. Find the gap in (1,0) cell (1 Mark)

Task 2: 1T-1R RRAM Array

a. Now create a 3 x 3 1T-1R Crossbar. You can use the same transistor settings we used in the Mid-Sem exam. (For writing, use a 1.2V pulse for 10ms to get an ON conductance of about 0.4nS) (1 Mark)

b. Follow the same steps, but now instead of the V/3 scheme, just turn the WL of the 2nd and 3rd rows off. Use the BLs of the 1st row to program (0,0) appropriately (1 Mark)

c. Now read the gap in (1,0) cell. Compare it with the previous part (2 Marks = 1+1)

Task 3: Row-by-Row writing

a. Now instead of writing cell-by-cell. Write the following weights row-by-row. Report the number of pulse-cycles taken to program this. (1 Mark)

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

b. Find the gaps of all the cells and report them (3 Marks = 1+1+1)